Detailed mappings for $\mu \in \{3, 4, 5\}$

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1 Supplementary Material: Detailed Layer Structures

This supplement provides explicit layer mappings, closed-form difference formulas, and product identities for $\mu \in \{2, 3, 4, 5\}$. These serve as concrete examples of the DSRS framework and validate the formulas in Equations (5)–(7) and (10) of the main text.

1.1 Structure for $\mu = 2$

Layer Mapping

n^2	1		4		9		16		25		36		49		64		81		100
$\Delta_U(n)$		1		3		3		5		5		7		7		9		9	
$U(n)$	1		2		5		8		13		18		25		32		41		50
$C(n)$			2				8				18				32				50
$L(n)$	0		2		4		8		12		18		24		32		40		50
$\Delta_L(n)$		2		2		4		4		6		6		8		8		10	

Common layers $C(n)$ occur when n is even.

Closed-Form Formulas

$$\Delta_U(n) = 2\left\lceil \frac{n}{2} \right\rceil + 1, \quad \Delta_L(n+1) = 2\left\lceil \frac{n+1}{2} \right\rceil. \tag{1}$$

Sequences: $\Delta_U(n) \in \{3, 3, 5, 5, 7, 7, \ldots\}$ (odd, doubled); $\Delta_L(n+1) \in \{2, 4, 4, 6, 6, \ldots\}$ (even, doubled).

Residuals

For odd n: $\alpha = -1$, $\beta = +1$, so $n^2 = 2U(n) - 1 = 2L(n) + 1$.

Product Identity

$$\pi \approx 2 \prod_{n=1}^{\infty} \frac{\lceil (n+1)/2 \rceil \cdot 2}{\lceil n/2 \rceil \cdot 2 + 1}. \tag{2}$$

This encodes the Wallis product via floor-ceiling arithmetic.

1.2 Structure for $\mu = 3$

Layer Mapping

n^2	1		4		9		16		25		36		49		64		81		100	
$\Delta_U(n)$		1		1		3		3		3		5		5		5		7		
$U(n)$	1		2		3		6		9		12		17		22		27		34	
$C(n)$					3						12						27			
$L(n)$	0		1		3		5		8		12		16		21		27		33	
$\Delta_L(n)$		1		2		2		3		4		4		5		6		6		

Common layers when $3 \mid n^2$ (i.e., $n \in \{3k\}$).

Closed-Form Formulas

$$\Delta_U(n) = 2\left\lceil \frac{n}{3} \right\rceil + 1, \quad \Delta_L(n+1) = \left\lceil \frac{2(n+1)}{3} \right\rceil. \tag{3}$$

Residuals

For $n^2 \not\equiv 0 \pmod 3$: $\alpha = -2$, $\beta = +1$, so $n^2 = 3U(n) - 2 = 3L(n) + 1$.

Product Identity

$$\pi \approx 2 \prod_{n=1}^{\infty} \frac{\lceil 2(n+1)/3 \rceil}{2\lceil n/3 \rceil + 1}. \tag{4}$$

1.3 Structure for $\mu = 4$

Layer Mapping

n^2	1		4		9		16		25		36		49		64		81		100	
$\Delta_U(n)$		0		2		1		3		2		4		3		5		4		6
$U(n)$	1		1		3		4		7		9		13		16		21		25	
$C(n)$			1				4				9				16				25	
$L(n)$	0		1		2		4		6		9		12		16		20		25	
$\Delta_L(n)$		1		1		2		2		3		3		4		4		5		5

Common layers when n is even.

Closed-Form Formulas

$$\Delta_U(n) = 2\left\lceil \frac{n-1}{2} \right\rceil - \left\lfloor \frac{n-1}{2} \right\rfloor, \quad \Delta_L(n) = \left\lceil \frac{n}{2} \right\rceil. \tag{5}$$

Note: Skip initial terms where $\Delta_U(n) = 0$ or $\Delta_L(n) = 0$.

Residuals

For $n^2 \not\equiv 0 \pmod 4$: $\alpha = -3$, $\beta = +1$, so $n^2 = 4U(n) - 3 = 4L(n) + 1$.

Product Identity

$$\prod_{n=1}^{\infty} \frac{\lceil (n+1)/2 \rceil}{2\lceil n/2 \rceil - \lfloor n/2 \rfloor} \approx 1. \tag{6}$$

Demonstrates convergence to unity for $\mu \equiv 0 \pmod 4$.

1.4 Structure for $\mu = 5$

Layer Mapping

n^2	1		4		9		16		25		36		49		64		81		100
$\Delta_U(n)$		0		1		2		1		3		2		3		4		3	
$U(n)$	1		1		2		4		5		8		10		13		17		20
$C(n)$									5										20
$L(n)$	0		0		1		3		5		7		9		12		16		20
$\Delta_L(n)$		0		1		2		2		2		2		3		4		4	

Common layers when $5 \mid n^2$ (i.e., $n \in \{5k\}$).

Closed-Form Formulas

$$\begin{aligned} \Delta_L(n) &= \left\lceil \frac{n-1}{5} \right\rceil + \left\lceil \frac{n}{5} \right\rceil, \\ \Delta_U(n) &= \left\lceil \frac{n}{5} \right\rceil + \left\lceil \frac{n-1}{5} \right\rceil - \left\lceil \frac{n-2}{5} \right\rceil + 2\left\lceil \frac{n-3}{5} \right\rceil - \left\lceil \frac{n-4}{5} \right\rceil. \end{aligned} \tag{7}$$

Note: Skip initial zeros.

Residuals

For $n^2 \not\equiv 0 \pmod 5$: $\alpha \in \{-4, -1\}$, $\beta \in \{1, 4\}$ (periodic cycle).

Product Identity

$$\pi \approx 2 \prod_{n=1}^{\infty} \frac{\lceil (n-1)/5 \rceil + \lceil n/5 \rceil}{\lceil n/5 \rceil + \lceil (n-1)/5 \rceil - \lceil (n-2)/5 \rceil + 2\lceil (n-3)/5 \rceil - \lceil (n-4)/5 \rceil}. \tag{8}$$

Additional Structures

Complete layer mappings, code, and data for all $\mu \in [2, 200]$ available at:
https://github.com/AST12212224/DSRS